

ChatGPT



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Abstract:

This report discusses ChatGPT, it talks about what the transformer is, the phases of GPT, the good and bad impact of ChatGPT in education. This report also mentions ChatGPT's memory and the difference between saved memory and chat history. It covers how users' privacy is violated and what the user could do to protect themselves. It goes on to cover the benefits of ChatGPT in mental health, but also how it could harm it, it then goes on to discuss how ChatGPT spreads lots of misinformation. It covers how ChatGPT impacts negatively on the planet with its water footprint, and it has a brief argument on whether ChatGPT has consciousness. Lastly, this paper also covers all the cool features you could use in ChatGPT.

Introduction:

ChatGPT (Generative Pretrained Transformer) is an AI chatbot made by OpenAI [1] and launched on 30th November 2022. It quickly gained everyone's attention going viral in five days on social media platforms, people started sharing online all the incredible things it could do, things like human-like conversations, planning, stories, writing code programs and many other things [2].

ChatGPT uses machine learning algorithms so that it can process and analyze data to generate outputs [4] which means, they absorb large amounts of data like books, articles, and webpages, to understand language then generate responses and conversations [5].

A general introduction to GPT and Transformers.

To really understand GPT, we got to understand the brains first: transformers. Lots of LLMS- Large Language Models- (like ChatGPT) use transformers which is a neural network architecture, that predicts the words that come next by absorbing large quantities of texts and looking for patterns in them- like how words and phrases resonate with each other [6].

Transformers consist of two main parts:

- The encoder; understanding the user's input.
- The decoder: generating an output (response) to the users' input.

Transformers are highly effective in language modelling. BERT (Bidirectional Encoder Representation of Transformers) and GPT (Generative Pretrained Transformers) were created after studies have shown that encoders and decoders can work individually, whereas BERT is a stack of encoders and GPT is a stack of decoders.

GPT had gone through a couple of versions, the first of which is GPT-1 and it was developed on 2018 by OpenAI, followed by GPT-2, and GPT 3, but since it was only accessible with an API, it wasn't popular. It all changed in 2022, when ChatGPT that was powered with GPT-3.5 was made, that made interaction with LLM much easier and more accessible [7].

Now that the basics of GPT and Transformers, lets dive into something more interesting; the impact of ChatGPT on education.

The impact of ChatGPT on education:

It is no secret that many students have been turning to ChatGPT, whether it's for homework, assignments or even exams. Educators (teachers, professor, etc.) are getting worried about the way it's been used, because unlike before, students aren't learning, studying or doing their homework themselves. Here's some examples on how ChatGPT could badly impact students:

- It could weaken critical thinking
- Reduces students' ability to learn independently
- Could harm students' original thoughts and their creativity

- Overtime it could lead to a lack of curiosity
- Students could be dishonest in their work
- ChatGPT could lead to laziness in learning or studying [8].

A survey conducted by best colleges in October 2023, has found that 43% of the students that participated have admitted to using AI tools-like ChatGPT. Nearly 50% of these students also confessed to use AI (like ChatGPT) for exams and assignments [9].

However, it's not all bad there is a brighter side if used responsibly and with good intentions for school, ChatGPT could also be a good tool in education, here are some of good ways ChatGPT could impact education:

- Makes education more accessible (translation, reading out loud for the blind etc.).
- Helps prepare mock exams (generates practice question for the student to practice).
- Personalized learning (ChatGPT could understand the user's learning style and help them study accordingly).
- Helps with homework [10].
- Clarify doubts a student might have on a certain topic.

In short, ChatGPT could definitely be used well in education, it just depends on how it's used. With that in mind let's explore ChatGPT's memory.

ChatGPT: memory

When you chat with ChatGPT (voice, texting, image generating) it stores important information like, what you like, what you dislike and interests so that it could respond in a way that's more personal to the user. You can control what it remembers or what it forgets by deleting the memories in settings or turning off memory.

There are two memory settings.

- **Saved memory:** this is where the memory of the things you asked ChatGPT to remember. For example, "Remember I am allergic to nuts when you recommend a baking recipe" This will automatically save/update to memory, and the next time ChatGPT will act accordingly, like in the earlier example it won't recommend recipes with nuts. Some things you say get automatically stored without you asking, like "I read every day to relax after college" this will also get stored, even when you didn't ask. You could also ask for a specific memory to be deleted, for example "delete memory about my nuts allergy" and ChatGPT will delete the memories. memories are stored in a different place called the notebook.
- **Chat History:** this is where the memory that ChatGPT stores from the same chats to make the responses more useful. Unlike saved memories, these information aren't stored in the memories, but rather in the chats, so if you open another chat, all the information in previous chat won't be remembered, as it's not stored in the memories.

If you like you can turn both or one of these off in your settings. Important thing to remember is that your information (saved in memory, in past conversations) is used by OpenAI so they could improve their models, to stop that just turn off “Improve the model for everyone” in settings [11].

Speaking of your information being used, let's now dive into privacy concerns with ChatGPT.

ChatGPT and privacy concerns

ChatGPT collects data, like what you type or say in the chat box. If you tell it your address, emails or anything else it stores the data. Your email address is stored by OpenAI when you make an account, and it collects your IP address, information about your device, information gathered by cookies, and your location whenever you use its service. Which is a lot of information. But what does OpenAI do with all this data?

It uses some of your data (memories, chat logs, conversations) to train GPT models in hopes of making it more accurate and usable. Your data is being stored by AI companies-like OpenAI because they are running out of data, that's high quality and available on the internet. They have considered using something called synthetic data-which is data that the AI models have created themselves. A big worry about using this type of data is that the AI models could get weird and strange, so training ChatGPT with synthetic data will also make it weird. So OpenAI resorted to using data provided by its users [12].

Why does this matter? First, this is your data that's being used, and secondly, they are not being clear that they're gathering your data. Italy data protection watchdog in 2024 had fined OpenAI 15 million euros (64,048,500 rupees) for collecting and using users' data to train the AI (ChatGPT) and for lack of age verification, which leads to kids being exposed to inappropriate AI generated content [13].

Your information could also be shared with a third party like with affiliates, law enforcement, and the government, this could be seen written in OpenAI privacy policy. Additionally, the more data a company collects the longer it'll be stored and delicate things like breach or hacking will expose all the information and data you decided to share, this is really dangerous and shouldn't be taken lightly as it could risk things like scams and identity theft. And this has happened before, here's some examples of how unsafe your data is when you share it with ChatGPT;

In March 2023 a huge data leak happened in OpenAI and lots of users' data was leaked and exposed, their names, addresses, emails, and credit card information were all exposed. Also in the past Google researchers have found ways to make ChatGPT hand over training data (the data OpenAI gets from users to train ChatGPT)-which includes people's name, addresses and phone numbers.

Moreover, there are concerns that your login credentials could be sold in the dark web, in 2023 about a quarter million of OpenAI logins were reportedly sold there [12].

ChatGPT also stores your data in a data bank, so anything you share risks going somewhere in a public domain [15]. **Always remember**, if you're not paying for the product, you are the product.

Now knowing all this how can we stay safe on ChatGPT and similar AI tools?

- Use the account-less version of ChatGPT: this comes with a disadvantage, as it won't store anything in the memory which would make it less personal and fun.
- Turn off "Improve the model for everyone" in settings, so your data won't be stored and used to train ChatGPT.
- Don't share anything you want to keep private, things like phone numbers, addresses, card number, and government ID etc. (besides, why would you share that?) [14].

Now that we've covered privacy concerns and how to protect your privacy and data, let's head on to something a little deeper, ChatGPT and its impact on mental health.

ChatGPT and mental health

Many people turn to ChatGPT when things get rough, they open up the app and vent, and ChatGPT has proven to have significant potential help in mental health care as it has many benefits, such as:

- Improve accessibility; ChatGPT could provide help 24/7, which makes it easier for users to use it any time, even outside office hours. It also lessens waiting time, as usually mental health services have a really long wait list, whereas ChatGPT offers immediate help. ChatGPT also provides accurate information about mental health, ways to cope, and it also helps raise awareness, and reduce misconceptions.
- Staying anonymous and lessens shame and judgement; users could talk to ChatGPT anonymously, which can reduce fear of seeking help, users could also interact at their own pace which makes seeking help easier to those who are hesitant or too shy to seek help from another person.
- Long time support; ChatGPT could provide lots of support and monitoring of mental health for a long time.[16]

As we can see there is lots of benefits from ChatGPT in this area, there's even a special feature in ChatGPT just for mental health, it's the "Mental Health GPT-ChatGPT", which highlights how seriously this area is being considered in AI.

However, there's also a bad side to ChatGPT in mental health-as there is for many things-which include:

- Depression: Because ChatGPT answers in a human-like way, there is a chance that people replace it for real life human interactions, that could lead to social isolation, which in turn is a huge factor for depression.

- Education: as mentioned earlier, students are relying on ChatGPT for homework, assignments and exams, which leads to lack of productivity and creativity, lack discipline, and laziness.[17]
- Not accurate in identifying and diagnosing mental problems: ChatGPT could give you accurate information about mental health, but it is not intended to diagnose any mental health problems, because that could lead to inaccuracy which then leads people to be misled, misdiagnosed and mistreated.
- Bias and inequity: because ChatGPT is trained on peoples data and data on the internet, there will be certain biases in the algorithm or data sources which may discriminate against certain populations, which can trigger strong emotions and memories for some people.[16]

Moving on from memories, lets now explore ChatGPT and misinformation

ChatGPT and misinformation

Many people have been using ChatGPT like google, as its much faster and easier to get the information we need. However, like google, ChatGPT isn't always reliable. If you would to ask it some questions and it didn't know the answer it would still confidently give you an incorrect answer. It would even sometimes argue against you if you wanted to correct it. But why does it do that?

Search engines like google, collect information databases then show the user the most popular links. ChatGPT also collect information databases, but it works differently, whereas search engines will show you exactly how its written, ChatGPT will give you your answers based series of many guesses and the results it finds. The like hood of ChatGPT finding and spreading misinformation is high as the internet consists of about 62% of misinformation and because it lacks the ability to reason and think, like humans could, it basically takes everything at face value and cannot distinguish between an opinions piece like a blog, and well a researched article.

So, if a lot of people ask similar questions, they'll get similar answers if not, the exactly same answer, and then misinformation will spread [18]. It goes without saying, information from ChatGPT should be checked before using, just like information from search engines, which may be difficult as it wont always generate the cite it got the information from, so you'll have to research manually.

A random fact, did you know ChatGPT has no access to information after 2021? So, some of the information you might be getting is out of date [21].

In addition to misinformation, ChatGPT also impacts the environment.

ChatGPT and the environment

This might not be obvious, but ChatGPT negatively impacts the environment, and by a lot. By estimation ChatGPT emits 8.4 tons (8500 kg) of carbon dioxide, which is more than double the amount of carbon dioxide an average person emits, which is 4 tons (4000kg)[19]. Research at university of California, Riverside done by Shaolei Ren and his team which included Pengfei Li and Jianyi Yang, and Mohammad A. Islam[20], stated that AI models like ChatGPT have a really high water footprint- which is the amount of water used to produce goods- ,the study revealed that Microsoft has used 700000 Liters of water when training ChatGPT-3's data centre[19], which is also the same amount if would produce 370 BMW cars[20]. Why so much water? Because when training, large amounts of energy is used, which produces heat and would need large amounts of water to cool down. ChatGPT also uses lots of water during the inference process, like when answering or generating texts. Did you know a simple conversation with about 50 questions consumes 500ml of water? That's the exact amount of water in some water bottles. And the fact ChatGPT has millions of users using it each day, means a large quantity of water is used daily. OpenAI has responded to the public's concerns about this by saying they're looking for different ways to use computing powers [19]

Lets get philosophical about the next part, ChatGPT and does it have a conscious?

ChatGPT and consciousness

Do you believe ChatGPT has some level of consciousness? A study by Dr. Clara Colombatto who's a psychology professor at UW- and her assistant Dr. Steve Fleming who's a professor of cognitive neuroscience at University College London, have conducted a survey of 300 U.S participants. They gathered information about their attitudes and behaviours towards ChatGPT, things like experiences and familiarities with AI, they've found that two thirds of the participants believe ChatGPT has a conscious. Colombatto has said it was really surprising that people have thought that way, as ChatGPT is mainly only texts and a voice with no face. She found that the more people used ChatGPT the more likely that think its conscious because people have come to trust it which, by the way, is exactly what innovators want, because if people trust their product they'll use it often.[22]

But is ChatGPT conscious?

ChatGPT is good at copying humans conversations and interactions, which makes it seem like its self-aware and conscious. But is it conscious? The short answer is no, although a few people argue that ChatGPT should be considered conscious. The co-founder of OpenAI, Ilya Sutskever has said the algorithms behind ChatGPT might be "slightly conscious" as he put it.

Stephen Wolfram who's a computer science has argued that no, ChatGPT isn't conscious in his website, he says it uses math to figure out what word to use in different contexts based on the data it's been trained on. And a lot of philosophers agree with him, stating that for one to be conscious is to have individual experiences, goals and beliefs.

The founding director of Florida Atlantic University's Centre for the Future Mind, Susan Schneider said that current chat box, like ChatGPT use human data and writings, so to test if

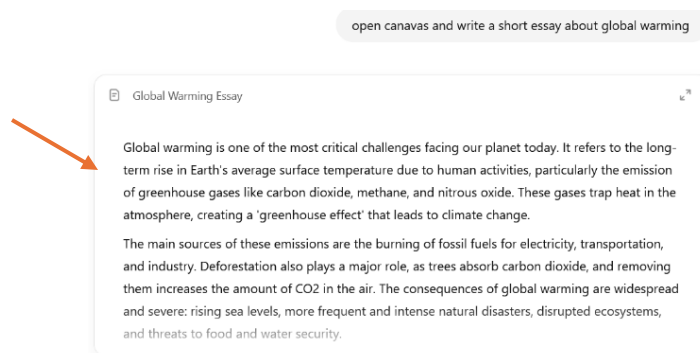
a program is conscious, they'd have to not give it access to any of that and see if it can describe its experiences. [23]

So, in short, while ChatGPT doesn't have a conscious, it does act like and interact like that, that said let's move on a different topic, features in ChatGPT.

Features in ChatGPT

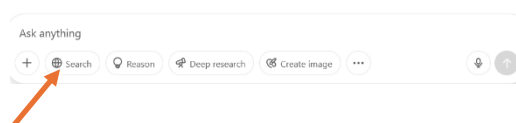
ChatGPT isn't just a Chat box, it has many cool features, here's a few; [25]

- **Canvas:** This is only available on Web, Windows, and MacOS, but they're working on making it available on mobile apps. Canvas is an interface that works with ChatGPT, and it helps with writing (essays, stories, etc) and coding your projects. Canvas helps ChatGPT understand the context and meaning of what you're trying to do. You can't access canvas directly on GPT-4 anymore, however ChatGPT will open a Canvas automatically when it understands what you're trying to do and needs it. For example, if you want to write an essay or program a short python code, ChatGPT will respond with a canvas. You could also use the prompt "open canvas..." to open a canvas manually [24].



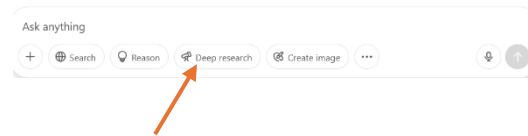
The white box is the canvas

- **Web browsing:** like stated earlier, ChatGPT has only been trained on data before 2021, so its knowledge is limited, however this feature was added to help ChatGPT browse the internet for more of the latest information to answer users' questions and to complete tasks. This can happen manually when ChatGPT senses it needs it or it needs more information or it can be done manually, if the user clicks on "search" you could also ask it for example, you could ask "Search about cool facts on seahorses" [25].

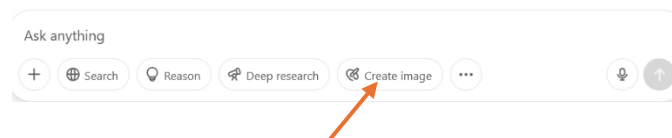


Deep searching: this extends normal searching, it does an in-depth multi source search using the data on the internet web, this allows ChatGPT to have a thorough and well

documented information, on the topics the user may've asked. [25] However there is a limit for free users, 5 times every 30 days.[26]

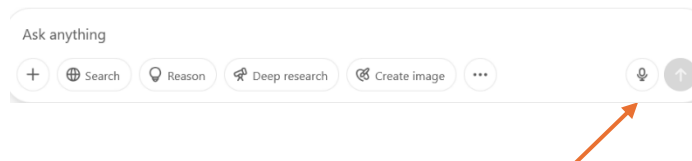


- **Image processing and generating** when a user uploads and image to ChatGPT, it uses GPT vision mode, it makes ChatGPT able to answer users' question about the image. With generating an image from the user's description, ChatGPT uses GPT-4o image generation model, and don't worry, your description doesn't need to be perfect. When the picture is generated, user could ask ChatGPT to edit it to [25]or they could ask it to edit an image from the user. To generate a picture the user could use "generate..." or they could click on "Create image" [27]



If the image ChatGPT has generated violates OpenAI usage policy, you can report it.

- **Voice mode:** you don't have to always text to ChatGPT, you could talk to it and it'll respond with a voice, and it doesn't sound robotic, it sounds with some pitch like humans. fortunately, this is available to mobile apps, desktop apps, on the website [25]. To access just press on the microphone icon.



There are many more features, but these are some examples.

In my opinion, ChatGPT is incredible and has so much great potential. I believe it could be extremely helpful in education as long as it used well, and it can help with mental health. However, I think people need to not get too comfortable with it and share everything as your data isn't safe, and it's best to use it only when you need to because its really bad for the environment and wastes a lot of water.

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